

U Shivakumar

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EDUCATION

PES University

Bachelor of Technology in Computer Science

Bangalore, India

Sept 2023 – Sept 2027

EXPERIENCE

Research Intern

June 2020 – Present

Center for Data Sciences and Applied Machine Learning (CDSAML), PES University

Bangalore, India

- Developed Alzheimer's Disease Classification using MRI scans from the ADNI dataset.
- Created YouTube tutorials on ADNI dataset access and preprocessing due to the lack of existing resources.
- Conducted comparative analysis of deep learning architectures (VGG, ResNet, DenseNet, EfficientNet-B2) for three-class classification: Cognitively Normal (CN), Mild cognitive Impairment (MCI), Alzheimer's Disease (AD).
- Implemented attention mechanisms including CBAM with ResNet-34 and Squeeze-and-Excitation blocks with EfficientNet-B2.
- Achieved 92% accuracy using EfficientNet-B2 with Squeeze-and-Excitation blocks.

Problem Solving With C - Teaching Assistant

Jan 2025 – May 2025

PES University

Bangalore, India

- Designed instructional slides and developed real-life application-based questions.
- Created practical programming exercises that bridge theoretical concepts with real-life scenarios.

Problem Solving With Python - Teaching Assistant

Aug 2025 – Jan 2026

PES University

Bangalore, India

- Assisted students during lab sessions for a class of 50+ students, providing one-on-one support to resolve doubts and clarify Python concepts.
- Guided students through problem solving approaches, and motivated them to develop logical thinking skills for coding challenges.

PROJECTS

PES Mentor Student Management System | *Agentic AI, Google ADK, BigQuery, MCP, Tavily, Google Cloud*

- Built an agentic AI-powered platform to help students discover mentors for academic guidance, research, and capstone projects.
- Enabled instant professor discovery using hybrid search (traditional filtering + vector embeddings) for semantic mentor recommendations.
- Integrated Google ADK with tools connected to BigQuery for scalable embedding-based retrieval.
- Utilized MCP with Tavily for deep external data retrieval (college websites, Google Scholar, LinkedIn) and automated summarization.
- Deployed the application on Google Cloud for real-time usage.
- Live: mentor-scout application

AI Automation Systems (n8n) | *n8n, Telegram API, Groq API, Gemini API*

- Developed an AI-powered Telegram shopping assistant with text and voice support for personalized product discovery.
- Integrated Groq API for audio transcription and built an agent to search products and respond interactively in real time.
- Reduced user effort in product search by automating comparison and recommendation workflows.
- Built an AI-based news summarization system to automate daily technology updates from multiple sources.
- Aggregated and summarized content using Gemini API and delivered structured insights via automated email newsletters.
- Implemented scheduled workflows (10 AM trigger) for consistent and reliable content delivery.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, SQL

AI/ML: Agentic AI, LLMs, RAG, Vector Embeddings

Frameworks: React, Node.js, Apps Script

Tools: Git, Docker, GCP, BigQuery, MCP, n8n